WORKED SOLUTION

STEP I 1987, Question 1

Pure • Curve Sketching

> **STEP I 1987, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/87-s1-q1

The key idea

Differentiating with the product rule leaves the factor e^{ax}, which is never zero, times a combination of $\cos bx$ and $\sin bx$. So the stationary points come from a single equation $\tan bx = \text{constant}$, whose solutions are equally spaced, a distance π/b apart. Moving from one to the next multiplies the exponential by a fixed factor and flips the sign of the cosine, and that is exactly a geometric progression.

Solution

Stationary points

By the product rule,

$$f'(x) = ae^{ax}\cos bx - be^{ax}\sin bx = e^{ax}(a\cos bx - b\sin bx).$$

Since $e^{ax} > 0$ for every x, we have $f'(x) = 0$ exactly when $a\cos bx = b\sin bx$.

> **Ask yourself** Can we safely divide by $\cos bx$ here?

Yes. If $\cos bx = 0$ then $\sin bx = \pm 1$, and the equation would say $0 = \pm b$, which is false because $b > 0$. So at any stationary point $\cos bx \neq 0$, and dividing gives

$$\tan bx = \frac{a}{b}.$$

Let $\alpha = \arctan(a/b)$. Because $a/b > 0$, we have $0 < \alpha < \frac{\pi}{2}$. As tan has period π, the solutions are $bx = \alpha + n\pi$, so the stationary points are at

$$x_n = \frac{\alpha + n\pi}{b}, \qquad n \in \mathbb{Z}, \qquad \text{where } \tan\alpha = \frac{a}{b}.$$

From a right-angled triangle with legs a and b, $\cos\alpha = \dfrac{b}{\sqrt{a^2+b^2}}$. Since $\cos(\theta + n\pi) =$

$(-1)^n \cos\theta$, the value of f there is

$$\begin{aligned} \mathrm{f}(x_n) &= \mathrm{e}^{a(\alpha+n\pi)/b}\cos(\alpha+n\pi) \\ &= (-1)^n \frac{b}{\sqrt{a^2+b^2}}\,\mathrm{e}^{a\alpha/b}\,\mathrm{e}^{na\pi/b}. \end{aligned}$$

Maximum or minimum? Differentiating again,

$$\mathrm{f}''(x) = \mathrm{e}^{ax}((a^2-b^2)\cos bx - 2ab\sin bx).$$

At a stationary point $b\sin bx = a\cos bx$, so $2ab\sin bx = 2a^2\cos bx$ and

$$\mathrm{f}''(x_n) = \mathrm{e}^{ax_n}\cos bx_n\,(a^2-b^2-2a^2) = -(a^2+b^2)\,\mathrm{f}(x_n).$$

So x_n is a maximum when $\mathrm{f}(x_n) > 0$ (n even) and a minimum when $\mathrm{f}(x_n) < 0$ (n odd). The stationary points alternate: maximum, minimum, maximum, ...

The geometric progression

Since $0 < \alpha < \pi$, we have $x_n > 0$ exactly when $n \geq 0$. So the stationary points with $x > 0$ are $x_0 < x_1 < x_2 < \cdots$, and for each $n \geq 0$

$$\frac{\mathrm{f}(x_{n+1})}{\mathrm{f}(x_n)} = \frac{\mathrm{e}^{a(\alpha+(n+1)\pi)/b}\cos(\alpha+(n+1)\pi)}{\mathrm{e}^{a(\alpha+n\pi)/b}\cos(\alpha+n\pi)} = \mathrm{e}^{a\pi/b}\cdot(-1) = -\mathrm{e}^{a\pi/b},$$

using $\cos(\theta+\pi) = -\cos\theta$ and $\cos(\alpha+n\pi) \neq 0$. The ratio doesn't depend on n, so the values $\mathrm{f}(x_0), \mathrm{f}(x_1), \ldots$ form a geometric progression with first term $\dfrac{b\,\mathrm{e}^{a\alpha/b}}{\sqrt{a^2+b^2}}$ and common ratio $-\mathrm{e}^{a\pi/b}$, as required.

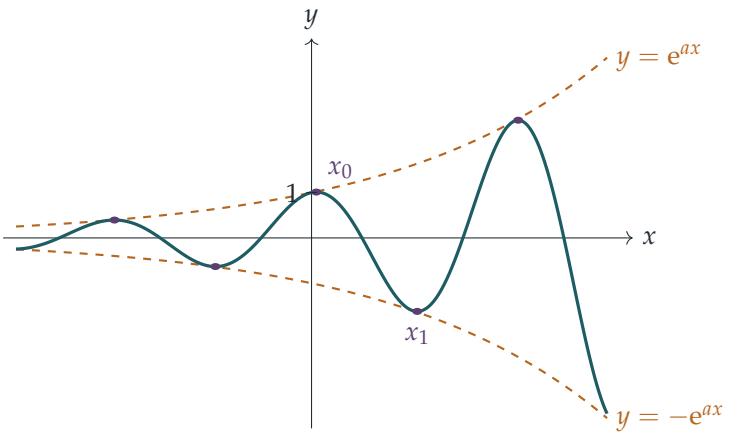
(The same calculation works for negative n too, so in fact all the stationary values form a single two-way geometric progression. The restriction to $x > 0$ just gives it a natural first term.)

Sketch

The facts we need:

- Since $-1 \leq \cos bx \leq 1$, the graph lies between the curves $y = \mathrm{e}^{ax}$ and $y = -\mathrm{e}^{ax}$. It touches $y = \mathrm{e}^{ax}$ where $\cos bx = 1$ ($bx = 2m\pi$) and $y = -\mathrm{e}^{ax}$ where $\cos bx = -1$ ($bx = (2m+1)\pi$). At these points $\sin bx = 0$, so $\mathrm{f}'(x) = \pm a\mathrm{e}^{ax}$, the same gradient as the envelope: the curve is tangent to it there, not turning.
- The zeros are at $bx = \frac{\pi}{2} + m\pi$, spaced π/b apart, and $\mathrm{f}(0) = 1$.
- Each turning point $bx = \alpha + n\pi$ lies just *after* the corresponding touching point $bx = n\pi$, because $0 < \alpha < \frac{\pi}{2}$: the rising envelope drags each peak slightly to the right.
- As $x \to \infty$ the oscillations grow without bound; as $x \to -\infty$, $\mathrm{e}^{ax} \to 0$, so $\mathrm{f}(x) \to 0$ while still oscillating.

The sketch shows $a = 0.3$, $b = 2$, with the envelopes $y = \pm e^{ax}$ dashed.



Checks

- Take $a = b = 1$: then $\alpha = \frac{\pi}{4}$, $x_0 = \frac{\pi}{4}$ and $f(x_0) = e^{\pi/4}/\sqrt{2} \approx 1.551$, while $f(x_1) = e^{5\pi/4} \cos \frac{5\pi}{4} \approx -35.89$. The ratio is $\approx -23.14 = -e^{\pi}$.
- As $a \to 0$ the function becomes $\cos bx$: $\alpha \to 0$, the turning points move to $bx = n\pi$, and the ratio tends to -1, giving the familiar $1, -1, 1, \ldots$
- Numerically, for 500 random pairs (a, b) we confirmed that $f'(x_n) = 0$, that $f(x_n)$ matches the formula above, that $f''(x_n) = -(a^2 + b^2)f(x_n)$, that consecutive ratios equal $-e^{a\pi/b}$, and that f' has no other zeros.

Common slips

- Putting the turning points where the curve touches the envelope, at $bx = n\pi$. Those points have gradient $\pm ae^{ax} \neq 0$; the turning points are a little to the right.
- Taking only $bx = \alpha$ and forgetting the general solution $bx = \alpha + n\pi$ of $\tan bx = a/b$.
- Missing the minus sign in the ratio: the cosine changes sign between consecutive stationary points, so the values alternate maximum, minimum.

Beyond STEP

The function $e^{ax} \cos bx$ is the real part of $e^{(a+ib)x}$, and functions like this are exactly the solutions of linear differential equations $y'' + py' + qy = 0$ whose auxiliary equation has complex roots. With $a < 0$ you get damped oscillation (a swinging pendulum with friction); with $a > 0$, as here, an unstable system whose oscillations grow. Successive peaks are in constant ratio, and the logarithm of that ratio is what engineers call the

logarithmic decrement. Boyce and DiPrima, *Elementary Differential Equations*, develops this in its chapter on second-order linear equations, and Needham, *Visual Complex Analysis*, shows why multiplying by $e^{(a+ib)x}$ traces a spiral.

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